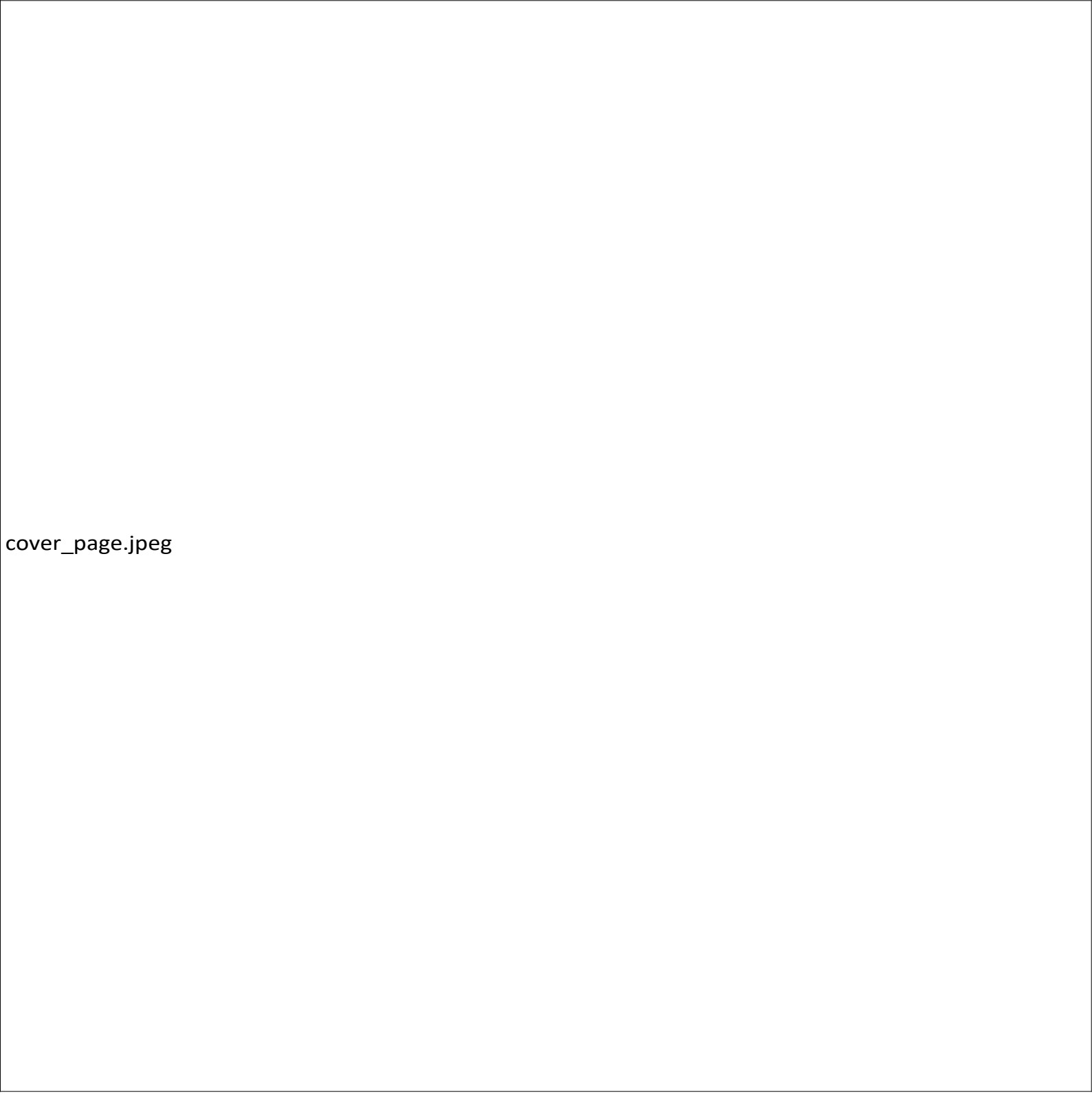


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Rizvi Education Society's

Rizvi College of Engineering

Department of Artificial Intelligence and Data Science Engineering

Project Synopsis Report

on

CampusVoice

NLP-Based Student Feedback Intelligence

Submitted in partial fulfilment of the requirements

of the degree of

Bachelors of Engineering

By

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2026 – 2027

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Certificate

This is to certify that the Mini Project Report entitled 'CampusVoice: NLP-Based Student Feedback Intelligence' has been submitted by Netra Chandrakant Neman (231A013)

under the guidance of Prof. Vijay Shankar

in partial fulfillment of the requirement for the award of the Degree of Bachelor of Engineering in Computer Engineering – Year 2026-27 Semester VII from University of Mumbai under the syllabus scheme R2019C.

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea, data, fact, or source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources that have not been properly cited or from whom proper permission has not been taken when needed.

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Date: _____

Abstract

CampusVoice is an NLP-based student feedback intelligence system designed to analyse natural language student comments and convert them into structured academic insights. The project presentation reports a dataset of 726 feedback records [7]. The processing pipeline includes text cleaning and normalization, explicit negation handling, word and character TF–IDF n-gram features, and Logistic Regression. The application predicts sentiment as Positive, Neutral, or Negative and also predicts a relevant academic category.

To make sentiment analysis more robust, CampusVoice combines the supervised model with VADER, a lexicon-based sentiment signal, and domain-specific rules. VADER is a rule-based sentiment analysis approach [1], while Logistic Regression and TF–IDF are supported by standard machine-learning tooling and literature [2–4]. The application returns a sentiment label, sentiment confidence, predicted academic category, category confidence, and information about negation or the analysis method.

The presentation reports sentiment accuracy of 84.14%, a sentiment F1 score of approximately 0.8435, and category accuracy of 66.44% [7]. A React interface and FastAPI backend provide the application layer for feedback analysis and dashboard-level summaries [5, 6]. The reported dashboard distribution is 62.26% Positive, 18.60% Neutral, and 19.15% Negative across 726 feedback records.

Keywords: Natural Language Processing, Student Feedback, Sentiment Analysis, TF–IDF, Logistic Regression, VADER, Negation Handling, FastAPI, React.

Index

Sr. No	Title	Page No
1	Introduction	1
2.	Problem Statement	2
2.1	Problem Statement	2
2.2	Existing Situation	2
2.3	Limitations of Existing Approaches	2
2.4	Scope	3
2.5	Expected Output	3
3.	Literature Survey	4
3.1.	Paper I	4
3.1.1	Overview	4
3.1.2	Research Gap and Methodology	4
3.1.3	Relevance to CampusVoice	4
3.2.	Paper II	5
3.2.1	Overview	5
3.2.2	Research Gap and Methodology	5
3.2.3	Relevance to CampusVoice	5
4.	Proposed System	6
4.1	System Overview	6
4.2	Methodology	6
4.3	Feature Engineering	6
4.4	Negation Example	7
4.5	Design Details	7
4.6	Prediction Confidence	7
5.	Implementation and Design	8
5.1	Technology Stack	8
5.2	Data Used	8
5.3	Application Output	8
5.4	Dashboard Design	9
5.5	Processing and Response Flow	9
5.6	Implementation Considerations	9
6.	Results and Evaluation	10
6.1	Reported Results	10
6.2	Evaluation Approach	10
6.3	Interpretation	11
6.4	Evaluation Limitations	11

7. Conclusion 13
7.1 Summary of Work 13
7.2 Key Contributions 13
7.3 Limitations 14
7.4 Future Scope 14
7.5 Final Remarks 14
References 15

List of Figures

Sr. No.	Title	Page No
4.1	CampusVoice NLP processing pipeline	6
4.2	High-level application architecture	7
6.4.1	Negative Feedback	11
6.4.2	Positive Feedback	12
6.4.3	Student Feedback Dashboard	12

List of Tables

Sr. No.	Title	Page No
2.5	Output Fields produced by CampusVoice	3
4.1	Negation-aware example from the project presentation	7
5.1	Technology and method summary	8
6.1	Results reported in the CampusVoice presentation	10
6.2	Sentiment distribution reported by the application	10

Chapter 1

Introduction

Educational institutions receive student feedback about teaching, laboratory facilities, learning resources, administration, and the overall academic experience. These comments are often written in natural language and may vary in length, spelling, tone, and wording. Reading every comment manually can be time-consuming, particularly when feedback is collected from many students.

Natural Language Processing (NLP) provides techniques for processing human language and extracting useful patterns from text. Sentiment analysis estimates whether a piece of text expresses a positive, neutral, or negative opinion. When sentiment labels are combined with an academic category, feedback can be grouped into more useful summaries.

CampusVoice applies an NLP pipeline to student comments. It cleans and normalizes text, considers negation, represents the text using word and character TF–IDF n-grams, and applies Logistic Regression. TF–IDF converts textual information into numerical features, and the project combines word-level and character-level representations [3]. A hybrid layer combines the model signal with VADER and domain-specific rules. The completed application is described as a React interface supported by a FastAPI backend [5], **Error! Reference source not found.**.

The project is motivated by the need to turn unstructured feedback into a form that can be reviewed more efficiently. A sentiment distribution provides a high-level view of feedback, while individual predictions and confidence values help users inspect the underlying comments. The system is intended to support interpretation rather than replace human judgement about student concerns.

Chapter 2

Problem Statement

2.1 Problem Statement

Student feedback is commonly collected as unstructured text. Extracting consistent sentiment and academic-category information from these comments can require manual reading and interpretation. The project addresses the problem of automatically processing student comments and presenting predicted sentiment and category information in a structured form.

2.2 Existing Situation

A collection of written comments may contain positive, neutral, and negative opinions, including words whose meaning depends on surrounding context. For example, a sentence containing the word "good" may express a negative opinion when it is preceded by "not." Feedback may also include spelling variation or strong emotional terms. These characteristics make simple keyword matching insufficient for all cases.

2.3 Limitations of Existing Approaches

The major limitations of CampusVoices are:

1. Process student feedback text through cleaning and normalization.
2. Detect explicit negation patterns so words are interpreted in context.
3. Represent text using word-level and character-level TF–IDF n-grams.
4. Use Logistic Regression to predict Positive, Neutral, or Negative sentiment.
5. Predict a relevant academic category and return confidence information.
6. Combine supervised prediction with VADER and domain-specific rules.
7. Present individual predictions and aggregate sentiment information through a React application backed by FastAPI.

2.4 Scope

The current project focuses on 726 feedback records and three sentiment classes: Positive, Neutral, and Negative. It also includes academic-category prediction, confidence reporting, negation information, and dashboard summaries. The supplied presentation does not provide a detailed breakdown of academic-category labels, the train/test split, or per-class evaluation results.

2.5 Expected Output

Table 2.5: Output fields produced by CampusVoice

Output field	Purpose
Sentiment label	Displays Positive, Neutral, or Negative.
Sentiment confidence	Shows the confidence associated with the sentiment prediction.
Predicted academic category	Groups the comment into a relevant academic category.
Category confidence	Shows confidence for the category prediction.
Negation/method information	Indicates whether negation was detected and/or which analysis method was used.

Chapter 3

Literature Review

3.1 Paper I: VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text

Authors: C. J. Hutto and Eric Gilbert (2014)

3.1.1 Overview

Sentiment analysis is an important Natural Language Processing (NLP) technique used to identify positive, negative, and neutral opinions expressed in text. Hutto and Gilbert developed VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon- and rule-based sentiment analysis model designed particularly for short social media texts. Their research addresses the difficulty of interpreting informal language, emotional expressions, and variations in sentiment intensity.

3.1.2 Research and Methodology

The authors developed and empirically validated a sentiment lexicon containing words associated with different sentiment intensities. They combined this lexicon with five general rules representing grammatical and syntactical patterns that influence sentiment, including emphasis and changes in sentiment intensity. They evaluated VADER against eleven existing sentiment analysis approaches, including lexicon-based methods and machine-learning techniques such as Naive Bayes, Maximum Entropy, and Support Vector Machines. Their experiments showed that VADER performed particularly well on social media text, reporting an F1 score of 0.96 for tweet sentiment classification

3.1.3 Relevance to CampusVoice

The research demonstrates that combining a sentiment lexicon with linguistic rules can improve sentiment interpretation without requiring a large supervised training process. In CampusVoice, VADER is used as an additional sentiment signal alongside Logistic Regression and education-specific rules to analyse student feedback. This supports the project's hybrid approach to identifying positive, neutral, and negative opinions. However, the results reported by Hutto and Gilbert apply to their evaluation datasets and should not be treated as the measured performance of CampusVoice.

3.2 Paper II: Scikit-learn: Machine Learning in Python

Authors: Fabian Pedregosa et al. (2011)

3.2.1 Overview

Machine learning provides techniques for building models that learn patterns from data and make predictions. Pedregosa et al. introduced Scikit-learn, an open-source Python machine-learning library designed to make machine-learning algorithms accessible through a consistent, well-documented interface. The library supports supervised and unsupervised learning, model selection, and data processing.

3.2.2 Research Gap and Methodology

The authors presented a unified software library that integrates a wide range of machine-learning algorithms for medium-scale problems. Their work emphasized ease of use, computational performance, consistent APIs, documentation, and minimal dependencies. Instead of proposing one specific prediction algorithm, the research focused on providing a reusable framework through which researchers and developers could implement and evaluate different machine-learning methods in Python.

3.2.3 Relevance to CampusVoice

The paper established the importance of a consistent and accessible framework for implementing practical machine-learning workflows. In CampusVoice, Scikit-learn-related techniques are used for text feature extraction with TF-IDF and sentiment classification using Logistic Regression. These components convert student comments into numerical representations and use the extracted patterns to predict sentiment and academic categories. The paper provides a software and methodological foundation for this implementation, rather than directly proposing a student-feedback sentiment analysis system.

Chapter 4

Proposed System

4.1 System Overview

CampusVoice follows a pipeline that receives a student comment, preprocesses the text, handles negation, extracts TF–IDF features, applies a machine-learning model, and returns a senti-ment label with confidence information. VADER and domain-specific rules provide additional sentiment signals. The application then exposes individual results and aggregated dashboard summaries.

Student feedback → Preprocessing and normalization → Negation-aware processing → Word and character TF–IDF → Logistic Regression → Label and confidence

Figure 4.1: CampusVoice NLP processing pipeline

4.2 Methodology

1. **Input:** Receive a raw student comment.
2. **Preprocessing:** Clean and normalize the text.
3. **Negation handling:** Identify explicit negation patterns such as “not good.”
4. **Feature extraction:** Create word and character TF–IDF n-gram features.
5. **Classification:** Apply Logistic Regression to predict the relevant label.
6. **Hybrid signal:** Incorporate VADER and domain-specific sentiment rules.
7. **Output:** Return sentiment, confidence, academic category, category confidence, and negation/method information.
8. **Aggregation:** Summarize predictions in the application dashboard.

4.3 Feature Engineering

The use of both word and character n-grams is a central feature-engineering choice in the project. Word n-grams represent terms and phrases, and character n-grams capture partial patterns that may be useful when feedback contains spelling variation. The two feature representations are combined before classification [3].

4.4 Negation Example

Table 4.1: Negation-aware example from the project presentation

Input comment	Reported interpretation
"The laboratory facilities are not good."	NEGATIVE; negation detected: YES

The example demonstrates why the system should consider surrounding context instead of treating a positive word in isolation

4.5 Design Details

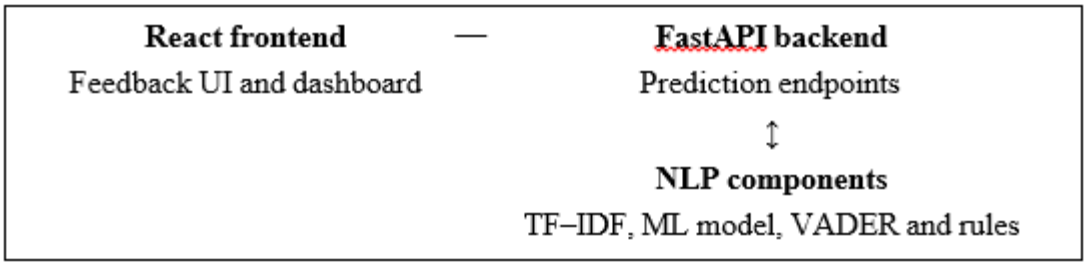


Figure 4.2: High-level application architecture

The presentation identifies React as the user-interface technology and FastAPI as the backend framework [5], **Error! Reference source not found.**. The diagram is a high-level representation of these stated components, not a detailed deployment diagram.

4.6 Prediction Confidence

The system returns confidence information for both sentiment and academic category. Confidence can help users distinguish predictions that the model supports more strongly from those that may need closer inspection. Confidence should not be treated as a guarantee that a prediction is correct.

Chapter 5

Implementation and Design

5.1 Technology Stack

Table 5.1: Technology and method summary

Component	Technology/method	Role
Frontend	React	User interface, feedback exploration, and dashboard.
Backend	FastAPI	Serves the NLP prediction workflow.
Text representation	Word and character TF–IDF n-grams	Converts text into numerical model features.
Sentiment classifier	Logistic Regression	Predicts sentiment labels and supplies model probabilities.
Lexicon signal	VADER	Adds a compound sentiment signal.
Domain logic	Education-specific rules and negation handling	Supports interpretation of domain terms and explicit negation.

5.2 Data Used

The presentation reports 726 feedback records. It does not specify the dataset collection method, demographic details, exact source-data label distribution, annotation process, or train/test split. These details should be added only after they are confirmed from the dataset and implementation.

5.3 Application Output

The application displays sentiment and the relevant academic category for a comment, together with confidence information and negation/method details. The presentation includes these example outputs:

- **NEGATIVE:** "The facilities are terrible and the equipment keeps breaking."
- **POSITIVE:** "The teacher explains everything very clearly and is extremely helpful."

These examples illustrate how the application labels individual feedback. They are examples from the supplied presentation and should not be interpreted as a complete test set.

5.4 Dashboard Design

The dashboard aggregates individual predictions into a campus-level summary. The supplied presentation reports the share of feedback classified as positive, neutral, and negative, alongside the total feedback count. Such a summary makes the overall distribution easier to review and can help identify areas where closer inspection may be useful.

5.5 Processing and Response Flow

The dashboard aggregates individual predictions into a campus-level summary. The supplied presentation reports the share of feedback classified as positive, neutral, and negative, alongside the total feedback count. Such a summary makes the overall distribution easier to review and can help identify areas where closer inspection may be useful.

5.6 Implementation Considerations

- Keep preprocessing consistent between training and inference.
- Ensure that negation handling is applied in a reproducible manner.
- Review class balance and label quality before extending the dataset.
- Expose confidence values without presenting them as certainty.
- Keep a human-review path for ambiguous or low-confidence feedback.

Chapter 6

Results and Evaluation

6.1 Reported Results

Table 6.1: Results reported in the CampusVoice presentation

Metric	Reported value	Interpretation
Sentiment accuracy	84.14%	Share of evaluated sentiment predictions reported as correct.
Sentiment F1 score	Approximately 0.8435	Reported F1 summary for sentiment classification.
Category accuracy	66.44%	Reported accuracy for academic-category classification.
Feedback records	726	Number of feedback records stated in the presentation.

The presentation states that sentiment prediction performs more strongly than category classification. The category model is more challenging, so the application presents category confidence along with the predicted category. Logistic Regression is a standard supervised learning approach [2], [4]. This report does not infer additional precision, recall, confusion-matrix, or per-class scores because those values were not included in the supplied material.

6.2 Evaluation Approach

Table 6.2: Sentiment distribution reported by the application

Sentiment	Reported share
Positive	62.26%
Neutral	18.60%
Negative	19.15%
Total records	726

The reported percentages sum to approximately 100% because of rounding. Positive feedback forms the largest share in the displayed dashboard, while neutral and negative feedback account for the remaining share. These percentages describe the application output shown in the presentation, not necessarily an independently audited ground-truth

distribution.

6.3 Interpretation

- The reported sentiment accuracy is 84.14%, with a reported F1 score of approximately 0.8435.
- The reported category accuracy is lower at 66.44%, indicating that academic-category prediction is a more difficult task in the current evaluation.
- The hybrid layer is described as an enhancement for unseen feedback; it should not be interpreted as a claim of perfect accuracy.
- Confidence values and individual comment inspection are important when using predictions to support academic review.

6.4 Evaluation Limitations

The supplied presentation does not include the train/test split, cross-validation setup, class-wise precision and recall, confusion matrix, baseline comparison, or a separate quantitative evaluation of the hybrid layer. Therefore, the reported metrics can be reproduced in this report, but the evaluation protocol cannot be independently assessed from the presentation alone. These details should be documented in a final technical version of the report.

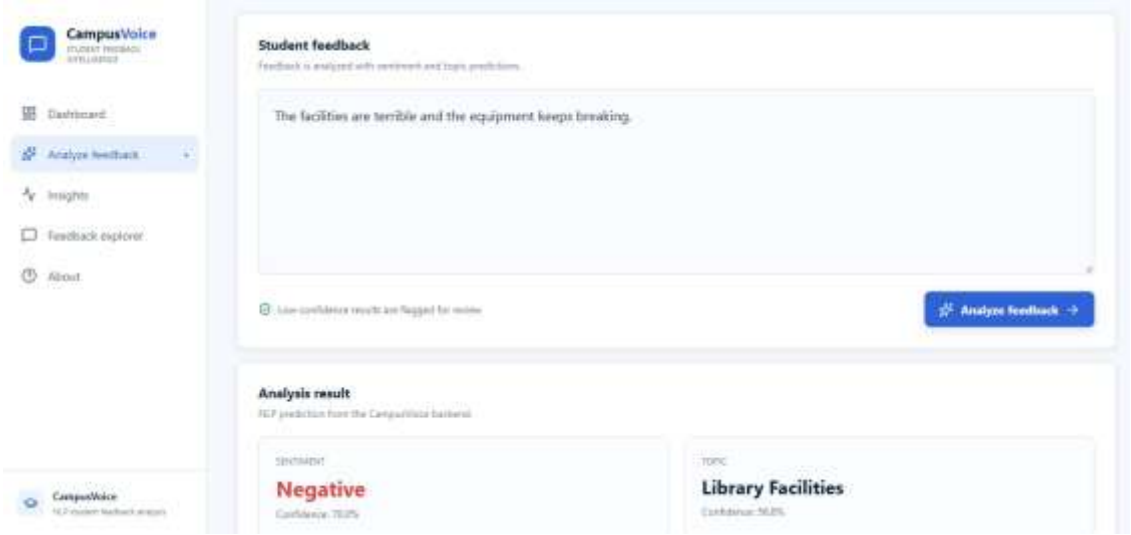


Figure 6.4.1:Negative Feedback

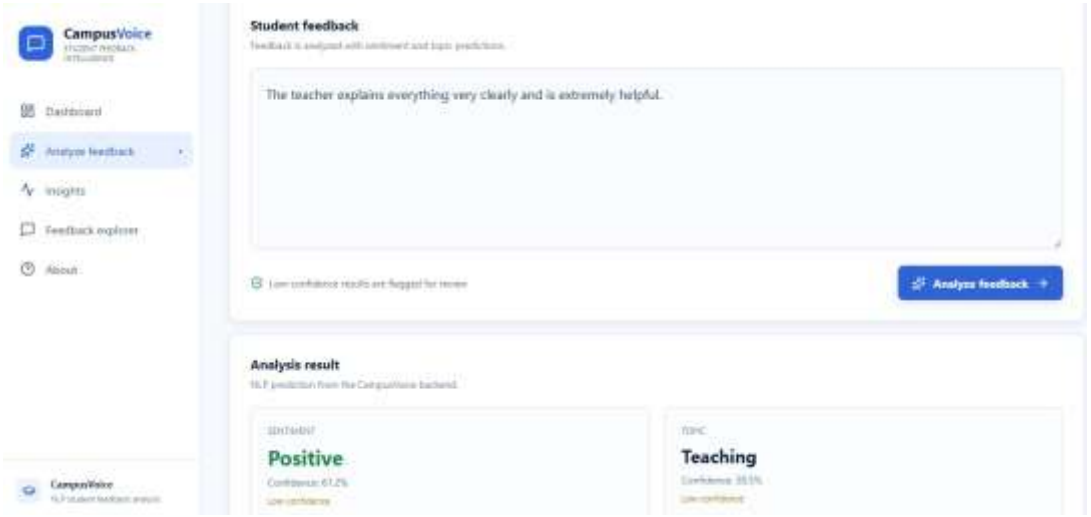


Figure 6.4.2: Positive Feedback

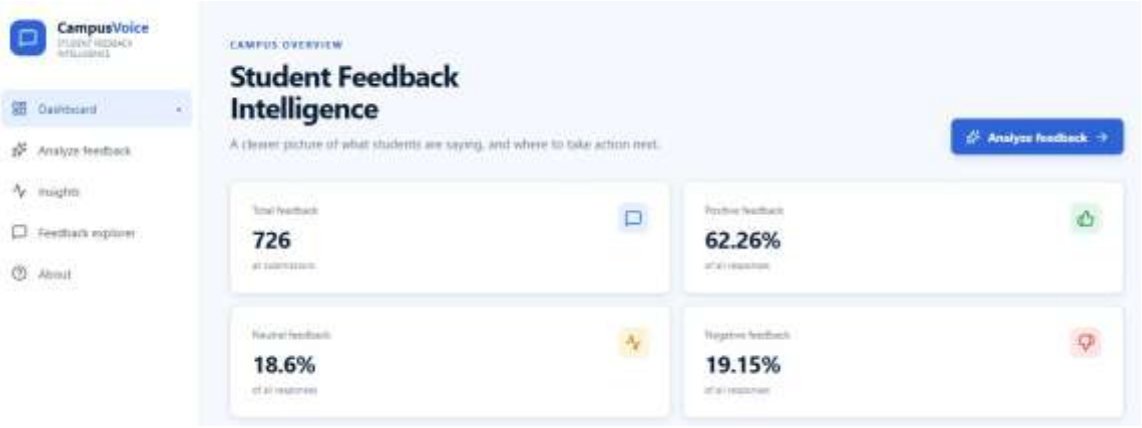


Figure 6.4.3: Student Feedback Dashboard

Conclusion

7.1 Summary of Work

CampusVoice demonstrates an end-to-end NLP workflow for transforming unstructured student comments into structured feedback insights. The system combines text cleaning and normalization, negation-aware processing, word and character TF–IDF features, Logistic Regression, VADER, and domain-specific rules. VADER provides a lexicon-based signal [1], while TF–IDF and Logistic Regression provide the feature representation and supervised classification components [3], [4]. The application predicts sentiment and academic category and provides confidence information through a React interface supported by FastAPI [5], **Error! Reference source not found.**.

The project presentation reports 84.14% sentiment accuracy, an approximate sentiment F1 score of 0.8435, and 66.44% category accuracy across 726 feedback records. The dash-board summarizes the reported predictions as 62.26% Positive, 18.60% Neutral, and 19.15% Negative. These results show the practical value of organizing feedback automatically, while also indicating that category classification and careful review remain important.

7.2 Key Contributions

- An end-to-end student-feedback processing pipeline.
- Negation-aware text handling to reduce misleading word-level interpretations.
- Combined word and character TF–IDF feature engineering.
- Logistic Regression models for sentiment and academic-category prediction.
- A hybrid sentiment layer using ML, VADER, and domain-specific rules.
- Confidence-aware output and a dashboard for aggregate feedback insights.

7.3 Limitations

- The dataset size reported in the presentation is 726 records; broader coverage may be needed for deployment across different institutions or student populations.
- Category accuracy (66.44%) is lower than sentiment accuracy (84.14%).
- The supplied presentation does not include class-wise metrics or a full evaluation protocol.
- The reported material does not quantify the incremental benefit of the hybrid layer separately.

7.4 Future Scope

- Expand the dataset with more diverse and representative feedback.
- Support multilingual sentiment analysis.
- Add aspect-based sentiment analysis to identify the specific issue discussed in a comment.
- Evaluate BERT or other transformer-based models.
- Provide semester-wise trend analysis.
- Detect recurring issues across feedback records.
- Introduce human-in-the-loop correction for low-confidence predictions.

7.6 Final Remarks

CampusVoice provides a practical foundation for analysing student feedback at scale. Further work should focus on broader data coverage, transparent evaluation, category-model improvement, privacy safeguards, and systematic review of uncertain predictions. With these improvements, the system can become a more reliable tool for understanding recurring academic feedback themes.

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